

SONGKHLA, Thailand

WMO: 485680

Lat: 7.187N

Lon: 100.608E

Elev: 4

StdP: 101.28

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 23.5	(c) 23.9	(d) 19.8	(e) 14.6	(f) 27.5	(g) 20.5	(h) 15.2	(i) 27.7	(j) 7.5	(k) 27.7	(l) 6.9	(m) 27.8	(n) 1.0	(o) 240	(p) 0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.5	34.2	25.9	33.5	26.0	33.0	26.1	27.5	31.8	27.2	31.5	26.9	31.2	3.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.7	30.3	26.0	21.4	30.1	25.7	21.0	29.7	87.4	32.0	86.0	31.6	84.7	31.5	32.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.4	5.5	5.1	DB	22.0	36.3	1.0	1.4	21.3	37.3	20.7	38.1	20.1	38.9	19.3	39.9	
			WB	20.4	28.9	1.2	1.1	19.6	29.6	18.9	30.3	18.2	30.9	17.4	31.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.3	27.6	28.3	29.1	29.4	29.2	29.0	28.9	28.6	27.9	27.2	27.1	(6)
	DBStd	1.26	0.96	0.89	0.92	0.89	0.90	0.98	0.99	1.08	0.91	0.98	1.08	1.10	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6677	536	494	566	572	601	576	588	586	557	555	517	530	(10)
	CDD18.3	3635	278	261	308	322	342	326	329	327	307	297	267	272	(11)
	CDH23.3	40860	2864	2832	3599	4032	4197	3850	3836	3734	3386	3112	2696	2723	(12)
	CDH26.7	14553	672	789	1293	1731	1820	1602	1557	1501	1250	1000	703	636	(13)
(14) Wind	WSAvg	1.9	2.7	2.7	2.3	1.8	1.6	1.6	1.7	1.7	1.7	1.5	1.7	2.3	(14)
(15) Precipitation	PrecAvg	2163	114	31	51	80	116	105	91	120	127	288	549	443	(15)
	PrecMax	3355	682	125	527	271	299	316	189	365	326	579	1334	1215	(16)
	PrecMin	1329	2	0	0	0	0	11	1	5	14	80	46	59	(17)
	PrecStd	478	127	34	79	63	56	60	47	78	67	113	245	257	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.9	33.0	34.0	34.7	34.7	34.7	34.9	34.0	33.2	31.9	31.2	(19)
		MCWB	25.7	25.4	26.1	27.0	26.3	25.8	25.5	25.4	25.7	25.7	26.1	26.0	(20)
	2%	DB	30.4	30.9	32.0	33.0	33.8	33.7	33.7	33.9	33.0	32.1	30.9	30.3	(21)
		MCWB	25.5	25.6	26.0	26.8	26.6	26.0	25.7	25.5	25.8	25.7	26.1	25.8	(22)
	5%	DB	29.8	30.2	31.3	32.3	33.0	32.9	32.9	32.9	32.2	31.2	30.2	29.9	(23)
		MCWB	25.2	25.3	25.9	26.5	26.5	26.1	25.7	25.6	25.8	25.9	25.9	25.7	(24)
	10%	DB	29.2	29.8	30.8	31.8	32.2	32.1	32.0	32.0	31.4	30.6	29.7	29.2	(25)
		MCWB	24.9	25.1	25.8	26.4	26.5	26.1	25.7	25.6	25.7	25.8	25.7	25.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.6	27.1	28.0	28.1	27.7	27.2	27.2	27.2	27.1	27.1	27.0	(27)
		MCDB	30.0	30.3	31.2	32.4	32.6	32.0	31.7	31.8	31.6	30.8	30.3	29.8	(28)
	2%	WB	26.1	26.1	26.6	27.4	27.5	27.1	26.7	26.7	26.7	26.6	26.6	26.4	(29)
		MCDB	29.4	30.0	30.8	31.8	32.0	31.6	31.3	31.3	31.1	30.4	29.8	29.3	(30)
	5%	WB	25.7	25.7	26.3	27.0	27.1	26.7	26.4	26.3	26.3	26.3	26.2	26.0	(31)
		MCDB	28.9	29.5	30.4	31.3	31.6	31.2	31.0	31.0	30.7	30.0	29.3	29.0	(32)
	10%	WB	25.3	25.4	26.0	26.7	26.8	26.4	26.1	26.0	26.0	26.0	25.9	25.7	(33)
		MCDB	28.5	29.1	30.1	30.9	31.1	30.8	30.8	30.7	30.4	29.7	29.0	28.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.7	5.4	5.9	6.5	6.7	6.8	7.0	6.5	5.8	4.6	4.1	(35)
		MCDBR	4.7	5.2	6.1	6.5	7.2	7.5	7.8	8.0	7.5	6.7	5.5	5.0	(36)
	5% WB	MCWBR	2.3	2.3	2.4	2.5	2.7	2.7	2.6	2.6	2.6	2.7	2.4	2.4	(37)
		MCWBR	4.4	5.0	5.6	6.2	6.6	6.8	7.0	7.2	6.8	6.2	5.2	4.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.467	0.468	0.476	0.461	0.447	0.454	0.460	0.456	0.447	0.447	0.458	0.470	(40)
		taud	2.281	2.258	2.261	2.312	2.375	2.383	2.358	2.351	2.373	2.379	2.345	2.291	(41)
	Ebn at Noon		851	861	854	853	847	830	829	846	866	868	853	839	(42)
		Edh at Noon	136	142	143	134	123	120	124	127	126	124	126	132	(43)
(44) All-Sky Solar Radiation	RadAvg		4.77	5.74	5.88	5.82	5.23	5.00	4.95	5.09	4.99	4.61	4.08	4.00	(44)
	RadStd		0.37	0.32	0.44	0.27	0.27	0.22	0.23	0.17	0.18	0.32	0.38	0.47	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+121
Regional (15 neighbors)	+0.19	N/A	N/A	+0.31	+0.17	+0.19	N/A	N/A	+88	+89

Nomenclature: See separate page